

CashBook

Product Requirements Document

Build a simple, real, working cash management app.

The app must include frontend, backend logic, database, authentication, file storage, permissions and deployment.

End goal

A user can sign in, create a cash book, add money in and money out, see the balance, upload receipts, search transactions, and share the book with another user.

Keep the product simple. Do not add payment gateways, UPI, bank sync, GST, payroll or AI.

1. What is a CashBook?

A CashBook is a simple place to record money coming in and money going out.

Example: A small office starts with ₹10,000. It spends ₹800 on fuel and receives ₹3,000 from a customer. The CashBook records both entries and shows the new balance: ₹12,200.

The two basic entry types

Feature	What the user can do	Why it matters
Cash In	Record money received.	Example: sales, client payment, refund, owner deposit.
Cash Out	Record money spent.	Example: food, fuel, salary, office expense, purchase.

What this project is not

This is not a banking app. The app does not actually send or receive money. It only records and organizes transactions.

- No UPI integration
- No payment gateway
- No bank connection
- No GST filing
- No payroll
- No AI features

2. Project Goal

Build a clean and usable CashBook application that works from login to deployment. The project is designed to teach the full flow of a modern app.

Feature	What the user can do	Why it matters
Frontend	Build pages, forms, tables, filters and dashboard.	Learn how users interact with the product.
Backend logic	Validate data, calculate balances and enforce rules.	Learn how the product behaves.
Database	Store users, books, transactions, categories and members.	Learn structured application data.
Authentication	Sign up, login and logout.	Learn user identity.
Authorization	Owner, Editor and Viewer permissions.	Learn access control.
Storage	Upload receipt images or PDFs.	Learn file storage.
Deployment	Publish the final app.	Learn how a finished product goes live.

3. Main User Flow

1. A new user signs up or logs in.
2. The user creates a Cash Book.
3. The user adds an opening balance if needed.
4. The user records Cash In and Cash Out entries.
5. The balance updates automatically.
6. The user can search, filter, edit or delete entries.
7. The user can attach a receipt to a transaction.
8. The owner can invite another user to the book.
9. The dashboard shows a simple summary.
10. The user can view a report for a selected date range.

4. Required Screens

Feature	What the user can do	Why it matters
Login / Sign Up	Email/password login. Google login is optional.	Entry to the app.
Home / Books	Show all Cash Books owned or shared with the user.	Choose where to work.
CashBook Dashboard	Balance, Cash In, Cash Out and recent transactions.	Quick overview.
Transactions	Full transaction list with search and filters.	Main working screen.
Add / Edit Transaction	Form for Cash In or Cash Out.	Create and update records.
Reports	Summary for a selected date range.	Understand the money flow.
Members	Invite people and set roles.	Collaboration and permissions.
Settings	Book name, currency, categories, opening balance.	Basic configuration.
Profile	Name, email and logout.	Basic account management.

5. Core Features

5.1 Authentication

- Sign up
- Login

- Logout
- Forgot password
- Keep the user logged in after refresh
- Show only data the logged-in user is allowed to access

5.2 Cash Books

- Create a Cash Book
- Give it a name
- Choose currency
- Set an optional opening balance
- Rename the book
- Archive or delete the book
- Create more than one book
- Open a book shared by another user

Example book names: Personal, Apta Office, Shop, Project Expenses.

5.3 Transactions

Every transaction must be either Cash In or Cash Out.

Feature	What the user can do	Why it matters
Amount	Required. Must be greater than zero.	₹2,500
Type	Required. Cash In or Cash Out.	Cash Out
Description	Short reason for the entry.	Office chair
Date	Required.	13 Aug 2026
Category	Required or selected from a list.	Office
Payment method	Optional simple label.	Cash / Bank / Card / Other
Note	Optional longer note.	Purchased for meeting room
Attachment	Optional image or PDF receipt.	receipt.pdf

5.4 Transaction actions

- Add
- View
- Edit
- Delete
- Search by text
- Filter by date range
- Filter by Cash In / Cash Out
- Filter by category
- Filter by payment method

- Sort newest first by default

5.5 Balance

The app must calculate the balance automatically. Users should not manually type the current balance.

$$\text{Current Balance} = \text{Opening Balance} + \text{Total Cash In} - \text{Total Cash Out}$$

If any transaction is edited or deleted, the totals and balance must update correctly.

5.6 Categories

- Provide default categories
- Allow the user to create a custom category
- Allow category editing
- Do not delete a category if it would break old transaction history

Suggested defaults: Sales, Salary, Food, Travel, Fuel, Office, Marketing, Purchase, Refund, Other.

5.7 Receipt Upload

- Allow image or PDF upload
- Store the file in cloud storage
- Save the file reference with the transaction
- Show the attachment when viewing the transaction
- Allow replacement or removal if the user has permission

6. Collaboration and Roles

A Cash Book can be used by more than one person. The owner controls who can access it.

Feature	What the user can do	Why it matters
Owner	Full access. Can manage the book, transactions and members.	Exactly one owner is enough for this project.
Editor	Can add and edit transactions. Can upload receipts.	Cannot manage ownership.
Viewer	Can only view the book, transactions and reports.	Cannot change data.

Rules

- A user must never see a book they do not own or belong to.
- Viewer actions must be blocked in the backend, not only hidden in the interface.
- Only Owner can invite or remove members.
- Owner cannot accidentally remove their own access without transferring ownership or deleting the book.

7. Dashboard

Keep the dashboard simple. It should answer four questions immediately: How much money is left? How much came in? How much went out? What happened recently?

- Current Balance
- Total Cash In for selected period
- Total Cash Out for selected period
- Recent transactions
- Simple Cash In vs Cash Out chart
- Simple expense-by-category chart

Charts are useful, but they are not more important than correct numbers.

8. Reports

Reports do not need accounting-level complexity. A basic useful report is enough.

- Select start date and end date
- Show opening balance for that period if possible
- Show total Cash In
- Show total Cash Out
- Show closing balance
- Show transaction list
- Show category summary

Optional after the main project works: export CSV, Excel or PDF.

9. Simple Data Structure

The exact database can differ by technology. The important part is that the data has clear relationships.

Feature	What the user can do	Why it matters
users	User account and profile data.	id, name, email
books	Each Cash Book.	id, name, ownerId, currency, openingBalance
book_members	Who can access each book.	bookId, userId, role
transactions	Money in and money out.	bookId, type, amount, date, categoryId, note
categories	Categories available inside a book.	bookId, name
attachments	Receipt or file information.	transactionId, fileUrl, fileName
activity_logs	Optional record of important actions.	bookId, userId, action, time

Relationship to understand

- One user can have many Cash Books.
- One Cash Book can have many transactions.
- One Cash Book can have many members.
- One Cash Book can have many categories.
- One transaction can have an attachment.

10. Product Rules

1. Amount cannot be zero or negative.
2. Every transaction must belong to one Cash Book.
3. Every transaction must have a date and type.
4. Only authorized users can read or change a book.
5. Balance must always come from stored data and calculations.
6. Deleting or editing a transaction must update all totals.
7. Uploaded files must belong to the correct transaction and book.
8. Empty states must be understandable. Example: “No transactions yet.”
9. Errors must be shown clearly. Do not fail silently.
10. The app must work properly on desktop and mobile.

11. Design Direction

The design should feel calm, clean and easy. This is a utility product, not a visual experiment.

- White or very light interface
- Strong readable text
- One clear accent color
- Simple cards and tables
- Consistent spacing
- Clear Cash In and Cash Out labels
- Do not overload the dashboard
- Use confirmation before destructive actions
- Responsive on mobile

A user should understand where to add a transaction within a few seconds.

12. Activity History

This is recommended because it teaches a useful backend pattern.

- User added a transaction
- User edited a transaction
- User deleted a transaction

- User invited a member
- User changed a member role

Example: “Nihal edited ₹1,200 Fuel expense · 10:42 AM”.

13. What Is Required vs Optional

Feature	What the user can do	Why it matters
Required	Auth, books, transactions, balance, categories, filters, storage, collaboration, permissions, dashboard, reports, responsive UI, deployment.	The project is incomplete without these.
Optional	Google login, activity log, export, dark mode, advanced charts, pagination.	Add only after the required product works.
Do not build	UPI, payment gateway, bank sync, GST, payroll, AI, Tally integration.	These are outside Project 1.

14. Suggested Build Order

1. Set up the project and database.
2. Build authentication.
3. Create the Cash Book data model.
4. Build create / view / edit book screens.
5. Build Cash In and Cash Out transactions.
6. Make balance and totals correct.
7. Add categories.
8. Add search, filters and sorting.
9. Add file upload.
10. Add collaboration and role permissions.
11. Build dashboard and reports.
12. Handle empty states, errors and validation.
13. Test mobile responsiveness.
14. Deploy the application.
15. Test the final deployed version with two different user accounts.

15. Final Acceptance Checklist

The project is considered complete only when the following works in the deployed app:

- ☐ A new user can create an account and log in.
- ☐ A user can create more than one Cash Book.

- ☐ A user can add Cash In and Cash Out.
- ☐ The current balance is always correct.
- ☐ A transaction can be edited and deleted.
- ☐ Search and filters work.
- ☐ Categories work.
- ☐ A receipt can be uploaded and viewed.
- ☐ An owner can invite another user.
- ☐ Editor and Viewer permissions work correctly.
- ☐ Unauthorized users cannot access another book through a direct URL or API call.
- ☐ Dashboard totals match the transactions.
- ☐ Reports work for a chosen date range.
- ☐ The app works on mobile and desktop.
- ☐ The app is deployed and usable from a public URL.

16. Final Demo

For the final review, demonstrate the product using two user accounts.

1. Login as the Owner.
2. Create a book called “Apta Demo”.
3. Add an opening balance of ₹10,000.
4. Add at least 5 Cash In and 8 Cash Out transactions.
5. Upload one receipt.
6. Show search and at least two filters.
7. Invite the second account as Editor and add a transaction from that account.
8. Change the second account to Viewer and prove that editing is blocked.
9. Return to the Owner account and show dashboard totals and report.
10. Open the deployed mobile view and show that the main flow still works.